

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme
2. Import the ten datasets from the `Ozone_TimeSeries` folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```
#1
#Read Ozone data
NC2010 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv",
                  stringsAsFactors = TRUE)
NC2011 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv",
                  stringsAsFactors = TRUE)
NC2012 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv",
                  stringsAsFactors = TRUE)
NC2013 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv",
                  stringsAsFactors = TRUE)
NC2014 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv",
                  stringsAsFactors = TRUE)
NC2015 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv",
```

```

stringsAsFactors = TRUE)
NC2016 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv",
stringsAsFactors = TRUE)
NC2017 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv",
stringsAsFactors = TRUE)
NC2018 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv",
stringsAsFactors = TRUE)
NC2019 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv",
stringsAsFactors = TRUE)
GaringerOzone <- rbind.data.frame(NC2010,NC2011,NC2012,
NC2013,NC2014,NC2015,NC2016,NC2017,NC2018,NC2019)

```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")

# 4
GaringerOzone <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq.Date(from=as.Date("2010/01/01"),
                                to=as.Date("2019/12/31"),
                                by="day"))

colnames(Days) <- "Date"

# 6
GaringerOzone <- left_join (Days,GaringerOzone)

```

```
## Joining with 'by = join_by(Date)'
```

```
dim(GaringerOzone)
```

```
## [1] 3652    3
```

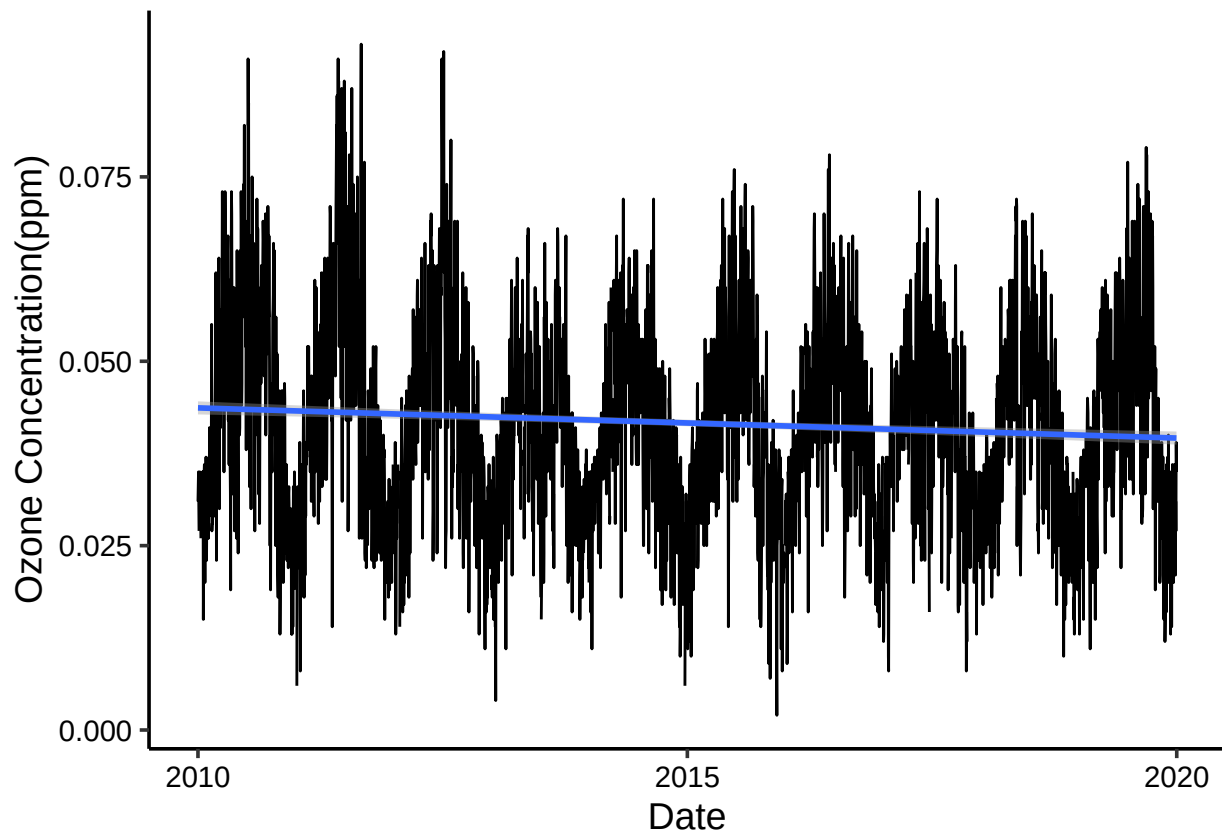
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7  
  
Ozone_plot<-  
ggplot(GaringerOzone, aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +  
  geom_line() +  
  ylab("Ozone Concentration(ppm)") +  
  geom_smooth( method = lm )  
print(Ozone_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```



Answer: The plot shows there is a negative trend over time even though the Ozone concentration showed strong seasonal peaks.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

```
GaringerOzone <-  
  GaringerOzone %>%  
  mutate(Daily.Max.8.hour.Ozone.Concentration =  
    zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration),  
    DAILY_AQI_VALUE = zoo::na.approx(DAILY_AQI_VALUE))
```

Answer: The piecewise is not for continuous data because Ozone data needs dynamic daily changes and also the spline interpolation only connect points which can harm the precision. So, using linear interpolation could solve these problems.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

#9

```
GaringerOzone.monthly <- GaringerOzone %>%  
  mutate(year=year(Date),  
    month=month(Date))%>%  
  group_by(year,month)%>%  
  summarise(mean_ozone=mean(Daily.Max.8.hour.Ozone.Concentration))
```

```
## 'summarise()' has grouped output by 'year'. You can override using the  
## '.groups' argument.
```

```
GaringerOzone.monthly <- GaringerOzone.monthly %>%  
  mutate(Date= my(paste0(month,"-",year)))  
  
head(GaringerOzone.monthly)
```

```
## # A tibble: 6 x 4  
## # Groups:   year [1]  
##   year month mean_ozone Date  
##   <dbl> <dbl>      <dbl> <date>  
## 1  2010     1      0.0305 2010-01-01  
## 2  2010     2      0.0345 2010-02-01  
## 3  2010     3      0.0446 2010-03-01  
## 4  2010     4      0.0556 2010-04-01  
## 5  2010     5      0.0466 2010-05-01  
## 6  2010     6      0.0576 2010-06-01
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

#10

#Daily observations

```
GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,  
                             start=c(2010,1),  
                             frequency=365)
```

#Monthly observations

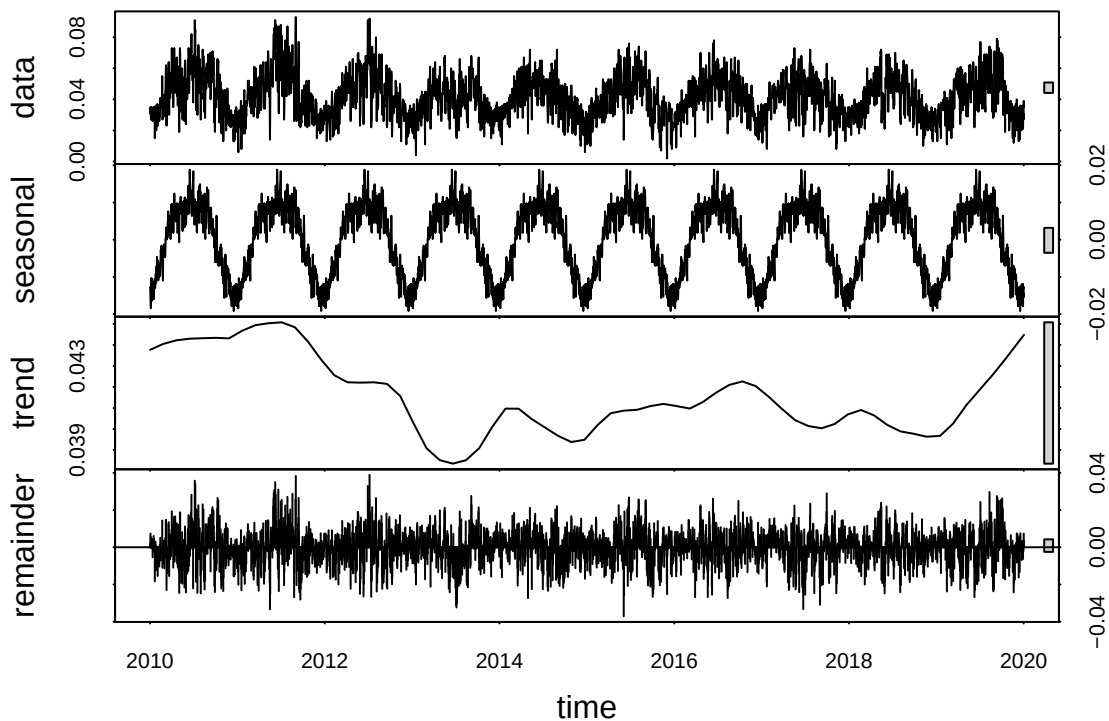
```
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$mean_ozone,  
                               start=c(2010,1),  
                               frequency=12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

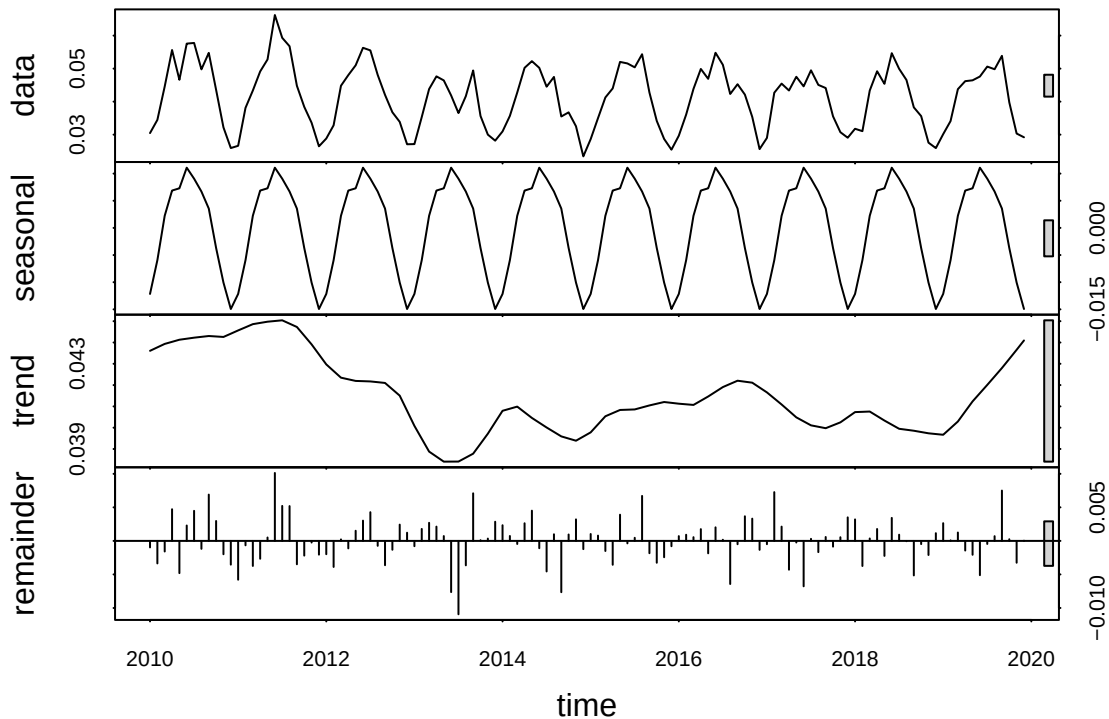
#11

#Decompose the daily time series

```
GaringerOzone.daily.ts_decomp <- stl(GaringerOzone.daily.ts ,s.window = "periodic")  
plot(GaringerOzone.daily.ts_decomp)
```



```
#Decompose the monthly time series
GaringerOzone.monthly.ts_decomp <- stl(GaringerOzone.monthly.ts,s.window = "periodic")
plot(GaringerOzone.monthly.ts_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

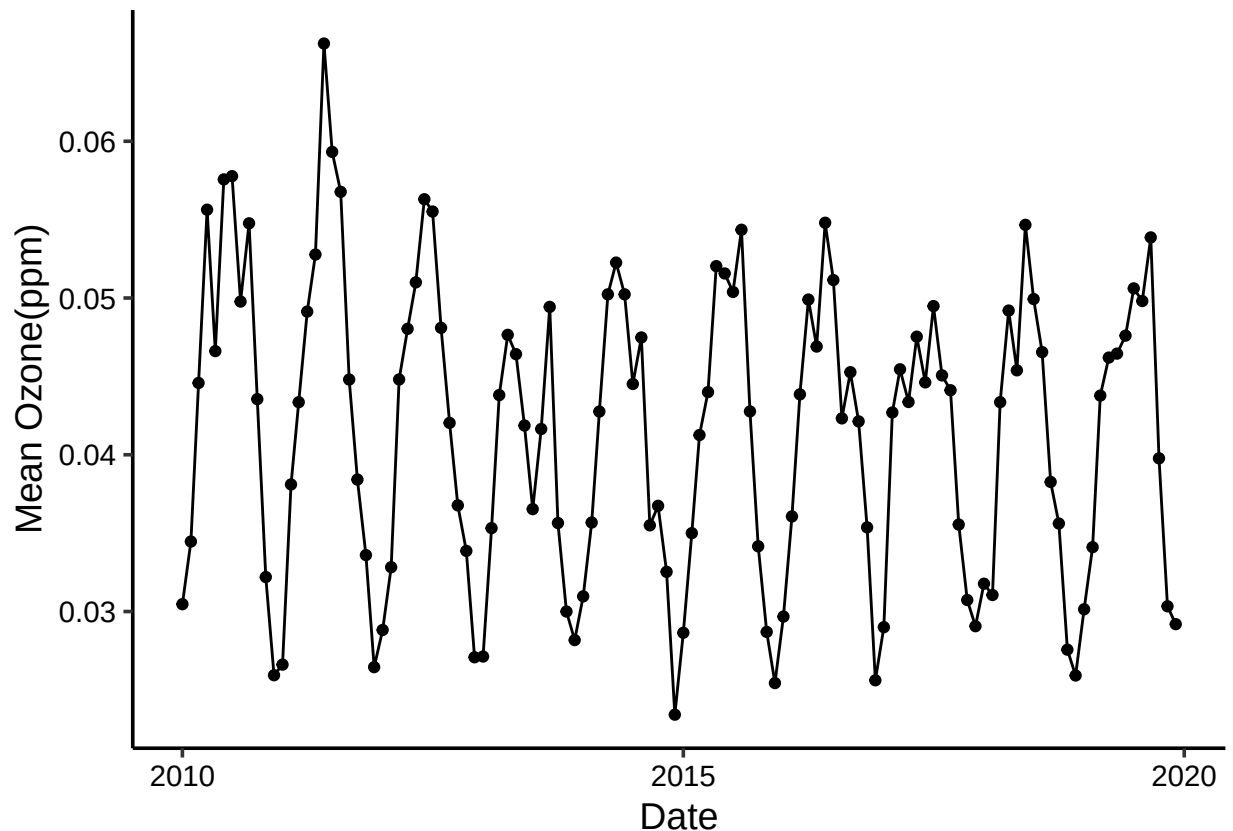
```
#12
# Run SMK test
GaringerOzone.monthly.ts_trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
print(GaringerOzone.monthly.ts_trend )
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: Our ozone data is likely not normally distributed, violating the normality assumption of parametric tests. Ozone data has strong seasonality, which violates the independence assumption. Therefore, we must use a non-parametric test that also accounts for seasonality, like the Seasonal Mann-Kendall test.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
#Visualization
monthly_ozone_plot <-
ggplot(GaringerOzone.monthly, aes(x =Date , y = mean_ozone)) +
  geom_point() +
  geom_line() +
  ylab("Mean Ozone(ppm)")
print(monthly_ozone_plot)
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: According to the mean monthly ozone concentrations over time, the analysis indicates that ozone concentrations have changed over the 2010s at this station with strong seasonality. The Seasonal Mann-Kendall test also shows a statistically significant decreasing trend in mean monthly ozone concentrations (Kendall's Tau = -0.143, p-value = 0.047).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```

#15
GaringerOzone.monthly.ts_Components <-
  as.data.frame((GaringerOzone.monthly.ts_decomp$time.series))

non_season_ozone <- mutate(GaringerOzone.monthly.ts_Components,
  Observed = GaringerOzone.monthly$mean_ozone,
  Date = GaringerOzone.monthly$Date)
non_season_ozone_fixed <- non_season_ozone$trend+non_season_ozone$remainder

#16
non_seasonal_trend <- Kendall::MannKendall(non_season_ozone_fixed)
print(non_seasonal_trend)

## tau = -0.165, 2-sided pvalue =0.0075402

```

Answer: Under the non seasonal Ozone monthly series, it also supports a statistically significant decreasing trend in mean monthly ozone concentrations (Kendall's Tau = -0.165, p-value = 0.008). We can conclude that both non-seasonal and seasonal mann kendall indicate monthly Ozone concentration has decreasing trend.